

Thomas Van Camp

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EDUCATION

University of Michigan, College of Engineering

Ann Arbor, MI

Major: Computer Science | Minor: Mathematics

August 2023 – May 2027

GPA: 3.94 / 4.0

- **Activities:** Tau Beta Pi (Engineering Honor Society), QuantUM Club (Quantum Science & Computing), Michigan Hackers Club (Security Team), Michigan Ski & Snowboard Club (President)
- **Honors:** James B. Angell Scholar, University Honors, Dean's List
- **Relevant Courses:** Architecting Hybrid Quantum-Classical Systems, Quantum Photonics, Quantum Nanotechnology, Intro to Quantum Computing (PHYS 380), Intro to Quantum Computing (EECS 479), Intro to Quantum Information Technology, Principles of Optics, Intro to Cryptography, Intro to Computer Security, Data Structures & Algorithms, Intro to Computer Organization, Foundations of Computer Science, Applied Linear Algebra, Discrete Math, Multivariable & Vector Calculus, Intro to Differential Equations, Intro to Probability, Theory of Algorithms

UNDERGRADUATE RESEARCH

University of Michigan, Computer Architecture for Quantum Advantage (CAFQA) Lab

Ann Arbor, MI

Principal Investigator Prof. Gokul Ravi, Computer Science & Engineering

August 2026 – Present

- Exploring and improving PureMagic, a Rust-based dynamic lattice surgery compiler for fault-tolerant quantum computing, with a focus on resource-efficient scheduling of logical operations and magic-state cultivation
- Investigating compiler scheduling and resource-allocation techniques for surface-code quantum computing
- Benchmarking tradeoffs between lattice surgery Pauli weight limits and operational parallelism

Michigan State University, Quantum Compilation & Simulation Lab (QCSL)

East Lansing, MI

Principal Investigator Prof. Ryan LaRose, Computational Mathematics, Science & Engineering

May 2026 – Present

- Selected as one of ten students to the NSF-sponsored MSU ACRES REU (Award #2349002)
- Conducted research on efficient classical simulations of large-scale unitary cluster Jastrow circuits
- Leveraged Qiskit SDK to create a pipeline to transform target molecules into quantum circuits, using PySCF for electronic structure calculations and Quimb for many-body physics and matrix product state (MPS) simulations
- Optimized scripts for GPU-based workflows on MSU's Institute for Cyber-Enabled Research (ICER) HPC cluster, utilizing ffsim for fermionic quantum circuit simulation and accelerating large-scale computational workflows
- Presented a poster and delivered an oral report of our research project findings at the 2026 Mid-SURE conference

PUBLICATIONS AND PRESENTATIONS

Hrishikesh Belagali, Thomas Van Camp, R. Pradeep, Sourin Das, Namit Anand, Ryan LaRose. "[Efficient classical simulation of large-scale unitary cluster Jastrow circuits.](#)" In review *Physical Review Letters* (2026). arXiv:2607.21337

Thomas Van Camp. "Classical Simulation of Large-Scale Unitary Cluster Jastrow (LUCJ) Circuits." Oral Presentation, Advanced Computational Research Experience Annual Symposium, Michigan State University, July 2026.

Thomas Van Camp, Hrishikesh Belagali, R. Pradeep, Sourin Das, Namit Anand, Ryan LaRose. "Classical Simulation of Large-Scale Unitary Cluster Jastrow (LUCJ) Circuits." Poster Presentation, Mid-Michigan Symposium for Undergraduate Research Experiences, Michigan State University, July 2026.

RELEVANT WORK EXPERIENCE

AbbVie Inc.

North Chicago, IL

Software Engineer Intern, Enterprise Artificial Intelligence Team

May 2025 – August 2025

- Developed a Chrome extension to create efficient and corporate-compliant social media content
- Collaborated with communications professionals to ensure accurate LLM responses and intuitive UX
- Developed a Streamlit web app for media reporting and summaries using web API endpoints
- Presented projects at several group meetings utilizing live demonstrations and posters
 - Technologies: GitHub, Python (Streamlit), LLMs, Chrome Extension APIs, Web UI/UX

AbbVie Inc.

North Chicago, IL

Software Engineer Intern, Therapeutics Artificial Intelligence Team

May 2024 – August 2024

- Created Flask APIs to auto-generate ten sections of Periodic Safety Update Reports via web requests
- Achieved 91% success rate for engineered LLM prompts to automate medical error narrative classification
 - Technologies: GitHub, Python, Flask RESTful, LLMs (GPT-4o, Claude, GPT-3.5 Turbo)

AbbVie Inc.

North Chicago, IL

Cloud Engineer Intern, Scientific and Cloud Computing Group

May 2023 – August 2023

- Created three scripts to maintain AWS cloud security and ensure compliance with updated corporate policies
- Deployed automated scripts via 200+ Jenkins pipelines to delete unapproved Identity Access Management (IAM) user keys, policies, and roles
 - Technologies: AWS IAM, AWS Cloud, Jenkins, GitHub, Python

AbbVie Inc.

North Chicago, IL

Cloud Engineer Intern, Scientific and Cloud Computing Group

May 2022 – August 2022

- Reduced AWS cloud vulnerabilities by 95% by creating automated scripts
- Deployed Ansible playbooks via Jenkins pipelines to run scripts and report vulnerabilities via daily emails
 - Technologies: AWS Cloud, Jenkins, GitHub, YAML, Ansible, Jira

AbbVie Inc.

North Chicago, IL

Cloud Engineer Intern, Scientific and Cloud Computing Group

May 2021 – August 2021

- Decreased AWS cloud costs by 90% through migration of CI/CD pipeline orchestration tools
- Used Dockerfile to create the Jenkins masters and agents, leveraging Ansible to deploy and run commands
- Implemented Jenkins pipelines in Amazon Elastic Kubernetes Service (EKS)
 - Technologies: AWS Cloud, AWS DevOps, Jenkins, Kubernetes, YAML, Docker, Ansible, Linux, Jira

PROJECTS

Benchmarking Quantum Error Correction (QEC) Decoders on the Surface Code

February 2026 – May 2026

- Designed and implemented a simulation pipeline for testing QEC decoders on the surface code
- Evaluated each QEC decoder's performance across accuracy, latency, threshold, robustness, and scalability tests
- Analyzed results of various physical error rates and code distances to determine practical use cases for each decoder
 - Source code and research report can be found at https://github.com/tvc04/qec_decoding

SHELLS

September 2025 – December 2025

- Created the game, SHELLS, using the Unity game engine, which was exhibited at public showcases and hosted online
- Designed and developed a 1v1 party-style game focused around a chaotic and synergistic upgrade system
- Conducted user acceptance testing (UAT) through multiple playtesting sessions

SKILLS

Languages: Python, C++, C, C#, Java, Rust, JavaScript, HTML, CSS, Swift, YAML

Tools: GitHub, LLMs, Qiskit, Streamlit, Slurm, Unity, Chrome Extensions, Web UI/UX, Pandas, Flask, AWS DevOps, AWS Cloud, Jenkins, Ansible, Docker, Kubernetes, Linux, Jira